

Lung and Colon Cancer Classification using a Custom Convolutional Neural Network

Abhinav Singh
EE25MT010

Neural Networks and Deep Learning
Indian Institute of Technology Dharwad, India
Email: ee25mt010@iitdh.ac.in

Abstract—Histopathological image analysis is important for cancer diagnosis. This paper introduces a custom Convolutional Neural Network (CNN) developed in PyTorch for the classification of lung and colon cancer histopathological images into five distinct subtypes: lung benign, lung adenocarcinoma, lung squamous cell carcinoma, colon adenocarcinoma, and colon benign. The model works with RGB images that have been resized to 128×128 pixels. It has three convolutional blocks and a three-layer fully connected classifier with dropout regularization. The network reached a test accuracy of 96.16% after 10 epochs of training with the Adam optimizer (learning rate 1e-4, weight decay 1e-4). This shows that it was good at extracting and generalizing features. The pipeline has data augmentation, validation-based checkpointing, and ways to see loss curves and confusion matrices. This research enhances computer-aided diagnostic systems for cancer detection.

I. INTRODUCTION

Lung and colon cancers are two of the most common types of cancer that kill people around the world. Early and precise diagnosis greatly enhances patient outcomes. Histopathological examination of tissue samples continues to be the gold standard, But doing it by hand takes a lot of time and is subjective. Deep learning, especially Convolutional Neural Networks (CNNs), has done very well at analyzing medical images [4]. CNNs learn hierarchical features from raw pixel data on their own. The lower layers find edges and textures, while the deeper layers find more complicated structures like nuclei and gland patterns. This capability obviates the necessity for manually crafted features and accommodates the significant variability inherent in histopathological images. We create and train a custom CNN from scratch using PyTorch [2] to sort histopathological images from the public Kaggle dataset [1] into five different types of cancer. The goal is to make a model that is accurate, efficient, and lightweight so that it can be used in environments with limited resources.

II. DATASET AND PREPROCESSING

A. Dataset Description

The dataset [1] contains 25,000 histopathological images of lung and colon tissues, each of size 256×256 pixels (RGB). The five classes are:

- Lung benign
- Lung adenocarcinoma
- Lung squamous cell carcinoma

- Colon adenocarcinoma
- Colon benign

The data was pre-split into Training, Validation, and Testing folders.

B. Preprocessing and Augmentation

To reduce computational cost and prevent overfitting, all images were resized to 128×128 pixels. For training, we applied:

- Random horizontal flip (probability 0.5)
- Random rotation (± 10 degrees)

Both training and test images were normalised using channel-wise mean and standard deviation of 0.5 (i.e., pixel values mapped from [0,1] to [-1,1]). No grayscale conversion was used; RGB information was preserved because colour can be a discriminating factor in stained tissue sections.

III. METHODOLOGY

A. CNN Architecture Overview

A CNN typically consists of alternating convolutional and pooling layers followed by fully connected layers. The convolutional layers apply learnable filters to extract spatial features. Batch normalisation accelerates training and improves stability. Pooling reduces spatial dimensions, making the model translation-invariant and reducing parameter count. Dropout randomly disables neurons during training to prevent co-adaptation and overfitting.

B. Model Architecture Details

Our model comprises a convolutional base and a fully connected (MLP) classifier.

TABLE I: Convolutional layers

Layer	Output size
Conv2d(3→16, 3×3, padding=1)	128×128×16
BN + ReLU + MaxPool(2)	64×64×16
Conv2d(16→32, 3×3, padding=1)	64×64×32
BN + ReLU + MaxPool(2)	32×32×32
Conv2d(32→64, 3×3, padding=1)	32×32×64
BN + ReLU + MaxPool(2)	16×16×64

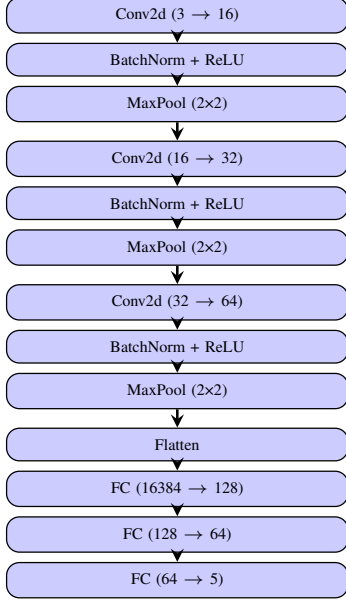
The number of filters doubles after each pooling step (16 → 32 → 64), a common design to increase representational capacity while reducing spatial size.

TABLE II: Fully connected layers

Layer	Output size
Flatten	16384
Linear + ReLU + Dropout(0.1)	128
Linear + ReLU + Dropout(0.1)	64
Linear	5

Total trainable parameters: approximately 2.1 million, which is modest compared to large pre-trained models, making it suitable for training on a single GPU or CPU.

Fig. 1: Visualisation of CNN blocks.



C. Training Configuration

- Loss function: CrossEntropyLoss (standard for multi-class classification)
- Optimizer: Adam with learning rate = 10^{-4} and weight decay = 10^{-4}
- Batch size: 64
- Epochs: 10
- Device: MPS (Apple Metal)
- Model checkpoint: saved whenever validation loss improves (best.pth)

Weight decay (L2 regularisation) penalises large weights, further reducing overfitting.

IV. RESULTS

Figure 2 shows the training and validation loss over 10 epochs. Both losses decrease steadily, indicating effective learning with minimal overfitting.

The final model achieved:

- Training accuracy: **96.97%**
- Testing accuracy: **96.16%**

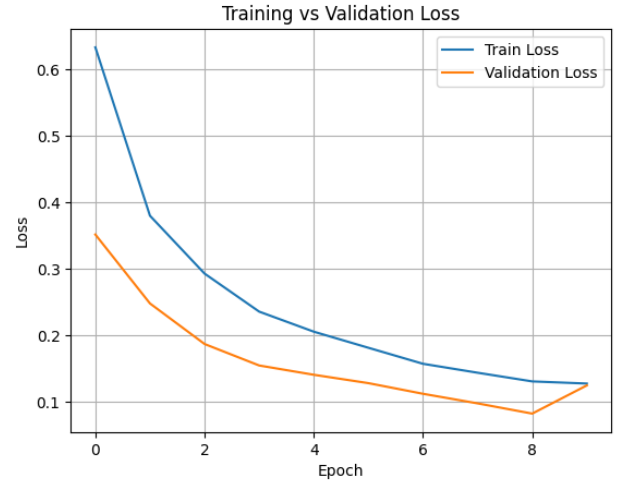


Fig. 2: Training and validation loss vs. epoch.

Figure 3 compares the two accuracies. The small gap (0.81%) suggests good generalisation, thanks to dropout and weight decay.

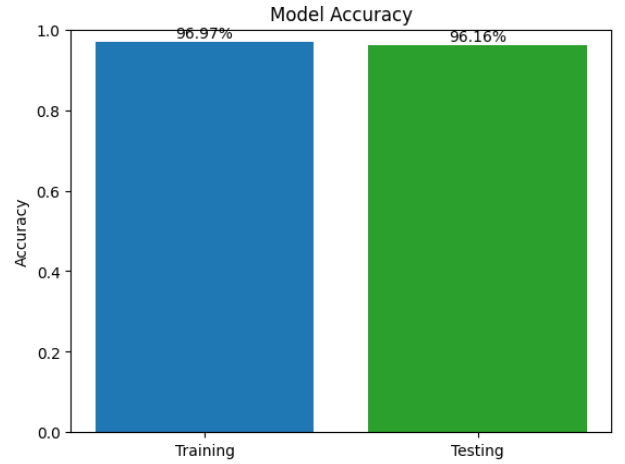


Fig. 3: Training vs. testing accuracy.

The confusion matrix (Fig. 4) provides per-class performance. Most classes are correctly classified, with minor confusion between some lung and colon subtypes – likely due to morphological similarities in certain tissue structures.

TABLE III: Classification report per class

Class	Precision	Recall	F1-Score	Support
colon_aca	0.98	0.99	0.99	1000
colon_n	0.99	0.98	0.99	1000
lung_aca	0.91	0.93	0.92	1000
lung_n	0.99	1.00	1.00	1000
lung_scc	0.93	0.91	0.92	1000
Accuracy		0.96		5000
Macro avg	0.96	0.96	0.96	5000
Weighted avg	0.96	0.96	0.96	5000

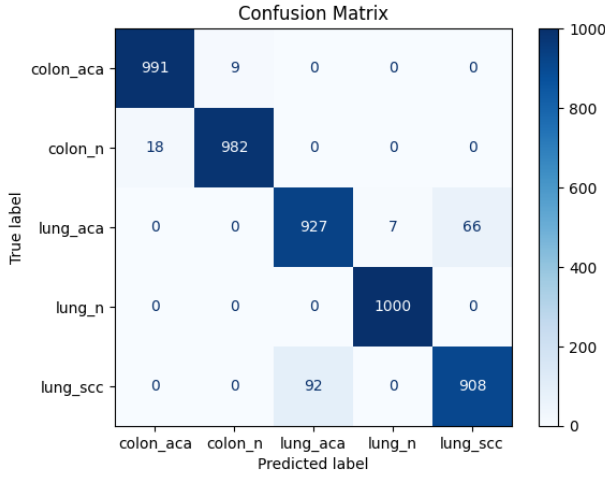


Fig. 4: Confusion matrix on the test set.

V. DISCUSSION

A. Observations

The results demonstrate that a modest CNN can achieve high accuracy (96.16%) on histopathological images. Key observations:

- Using RGB images (not grayscale) preserves colour information, which may help differentiate tissue types (e.g., haematoxylin and eosin staining variations).
- Dropout (0.3) and weight decay (10^{-4}) effectively reduce overfitting, as seen by the small gap between training and test accuracy.
- The best model was correctly saved and loaded for final evaluation, preventing overfitting to the last epoch.

B. Future Work

While the current model performs well, several improvements can be explored:

- **Longer training with early stopping:** Increase the number of epochs and monitor validation loss to stop when it plateaus.
- **Learning rate scheduling:** Use ReduceLROnPlateau to lower the learning rate when validation loss stagnates.
- **Stronger data augmentation:** Apply colour jitter, elastic deformations, or random erasing to improve robustness.
- **Transfer learning:** Fine-tune pre-trained models (ResNet50, EfficientNetB0) which often achieve higher accuracy with less training data.
- **External validation:** Evaluate on independent datasets to assess generalisation across different staining protocols and scanners.
- **Explainability:** Use Grad-CAM or SHAP to visualise which image regions influence predictions, building clinician trust.

VI. CONCLUSION

We built a custom CNN from the ground up using PyTorch to classify histopathological images into five classes. The

model got 96.16% accuracy on the Lung and Colon Cancer test set. The whole pipeline includes

loading data, adding to it, training with checkpoints, and showing loss curves and confusion matrices. This work helps make computer-aided diagnostic systems better for finding cancer. Future developments will concentrate on transfer learning and explainability to improve clinical relevance.

ACKNOWLEDGMENT

The author sincerely thanks the faculty of the **Neural Networks and Deep Learning** course at IIT Dharwad for their valuable guidance, insightful lectures, and continuous support throughout this project. Their emphasis on hands-on implementation and theoretical foundations made this work possible.

REFERENCES

- [1] A. M., "Lung and Colon Cancer Histopathological Images," Kaggle, 2020. [Online]. Available: <https://www.kaggle.com/datasets/andrewmvd/lung-and-colon-cancer-histopathological-images>
- [2] A. Paszke et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," in *Advances in Neural Information Processing Systems* 32, 2019.
- [3] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," in *Advances in Neural Information Processing Systems* 25, 2012.
- [4] G. Litjens et al., "A Survey on Deep Learning in Medical Image Analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.